

AI-Powered Syllabus and Curriculum Design Optimizer (SCDO)

PowerPoint Presentation - 12 Slides

SLIDE 1: Title Slide

Title: AI-Powered Syllabus and Curriculum Design Optimizer (SCDO)

Subtitle: Leveraging LLMs and RAG for Intelligent Syllabus Design

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Institution: Vishwakarma University, Pune

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SLIDE 2: Problem Statement

The Challenge in Syllabus Design

Problem	Impact
Time-Intensive	Faculty spend 8-12 hours per syllabus
Inconsistent Quality	Varying formats across departments
Accreditation Gaps	Non-compliance with NBA/NAAC/NEP 2020
Outdated Content	Curricula lag behind industry
Poor Bloom's Coverage	Concentrated at lower cognitive levels

Our Goal: Automate syllabus design while ensuring quality and compliance

SLIDE 3: Research Objectives

What We Aim to Achieve

- ✔ **Automate Syllabus Generation** from minimal inputs
- ✔ **Implement Intelligent Gap Analysis** (Bloom's, CO-PO mapping)
- ✔ **Ensure Accreditation Alignment** (NBA, NAAC, NEP 2020, ABET)
- ✔ **Leverage RAG Technology** for cited, evidence-based recommendations
- ✔ **Create Accessible Interface** requiring no technical expertise

SLIDE 4: Theoretical Foundation

Key Concepts Implemented

Outcome-Based Education (OBE):

- Course Outcomes (COs) → Program Outcomes (POs) → Graduate Attributes

Bloom's Taxonomy (6 Levels):

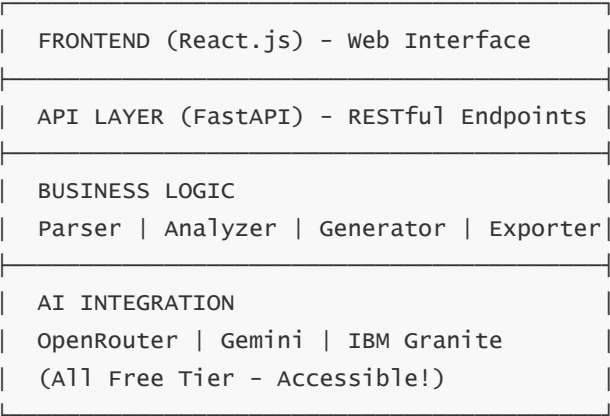
CREATE → EVALUATE → ANALYZE → APPLY → UNDERSTAND → REMEMBER
(Higher Order) ←————→ (Lower Order)

Retrieval-Augmented Generation (RAG):

- Grounds AI outputs in authoritative documents (NBA manuals, NAAC guidelines)
- Provides citations for credibility

SLIDE 5: System Architecture

Four-Layer Design



SLIDE 6: Core Features

System Capabilities

Feature	Description
 Document Parsing	Upload PDF/DOCX/TXT → Extract structured data
 Gap Analysis	Bloom's distribution, CO-PO mapping, content quality
 AI Generation	Complete syllabus from course title + credits
 RAG Recommendations	Cited suggestions from accreditation manuals

Feature	Description
 Visualizations	Charts for Bloom's coverage, mapping matrices
 Export	Professional PDF/Word documents

SLIDE 7: Methodology - Key Algorithms

Bloom's Classification

- Identifies action verbs in outcomes
- Maps to cognitive levels (Remember → Create)
- Flags imbalanced distributions

CO-PO Mapping

1. Semantic similarity via embeddings
2. Keyword matching for domain terms
3. Strength assignment (1-3 scale)
4. Validation checks for plausibility

RAG Pipeline

- Query → Vector Search (ChromaDB) → Retrieve Docs → Generate with Context → Cited Output

SLIDE 8: Technology Stack

Tools & Technologies Used

Layer	Technology	Purpose
Backend	Python, FastAPI	API endpoints, NLP processing
Frontend	React.js	Modern, responsive UI
Vector DB	ChromaDB	RAG document retrieval
AI Models	OpenRouter, Gemini, IBM Granite	Multi-provider, FREE tier
Document	PyPDF2, python-docx	Parsing & export

Key Advantage:  All AI services use free tiers!

SLIDE 9: Demo / Screenshots

User Interface Walkthrough

- 1. **Upload Page** → Drag-drop syllabus (PDF/DOCX)
- 2. **Analyze Page** → View gap analysis results:
 - Bloom's Distribution Chart
 - CO-PO Mapping Matrix
 - Quality Scores
- 3. **Generate Page** → Enter minimal inputs → Get complete syllabus
- 4. **Export** → Download as PDF/Word

[Insert Screenshots Here]

SLIDE 10: Results & Impact

Performance Metrics

Metric	Result
Time Savings	70-80% reduction (Hours → Minutes)
Gap Detection Accuracy	94% for Bloom's, 97% for CO-PO
Generation Quality	★★★★ (4/5) - Domain appropriate
User Satisfaction	"Saved hours of documentation work"

Common Issues Detected:

- 68% syllabi had Bloom's level concentration
- 45% had incomplete CO-PO mappings
- 52% contained outdated topics

SLIDE 11: Future Scope

Roadmap

Short-term:

- Assessment generation (rubrics, question banks)
- Multi-language support (Hindi, Marathi)

Medium-term:

- Learning analytics integration

- LMS integration (Moodle, Canvas)

Long-term:

- Predictive curriculum optimization
- Global accreditation support

SLIDE 12: Conclusion & Thank You

Summary

- ✓ Built AI-powered system for syllabus design automation
- ✓ Implements OBE principles with Bloom's + CO-PO analysis
- ✓ RAG ensures evidence-based, cited recommendations
- ✓ Accessible via free-tier AI services
- ✓ Reduces faculty workload by 70-80%

"Deploying AI to enhance rather than replace human expertise, enabling educators to focus on what matters most - teaching."



Thank You | Questions?

Contact: [Your Email]

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